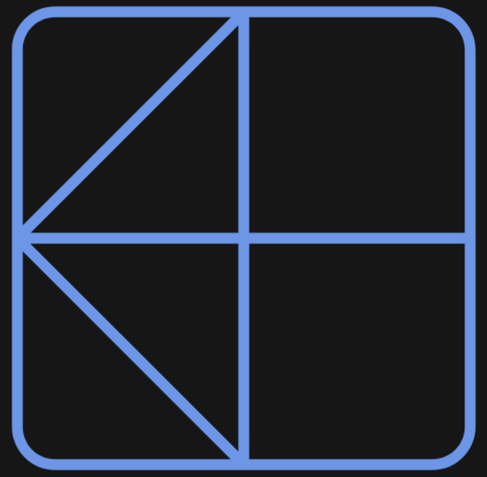




VitalWatch

Real-time AI for critical care decisions on IBM LinuxONE

IBM Z Datathon 2026 · Event code SSF2026 · Track: Real-Time AI for Critical Decisions · AI for Good

Sepsis needs earlier warning, and ICU teams already ignore most alarms

11 million

deaths from sepsis worldwide in one year

Rudd et al., Lancet 2020

+4%

higher odds of dying in hospital for every hour sepsis care is delayed

Seymour et al., NEJM 2017

72–99%

of clinical alarms are false, so staff learn to tune them out

Sendelbach and Funk, AACN 2013

Clinicians need warnings that are **earlier**, **fewer** and **explained**.

THE SOLUTION

Earlier, fewer, explained alerts, without exposing patient data



Real time on IBM Z

Every bed scored each hour in one call, on the IBM Telum AI accelerator through IBM Snap ML.



Fewer alarms

Alerts need sustained or sharply rising risk; snooze after a response; escalate if nobody responds.



Explained

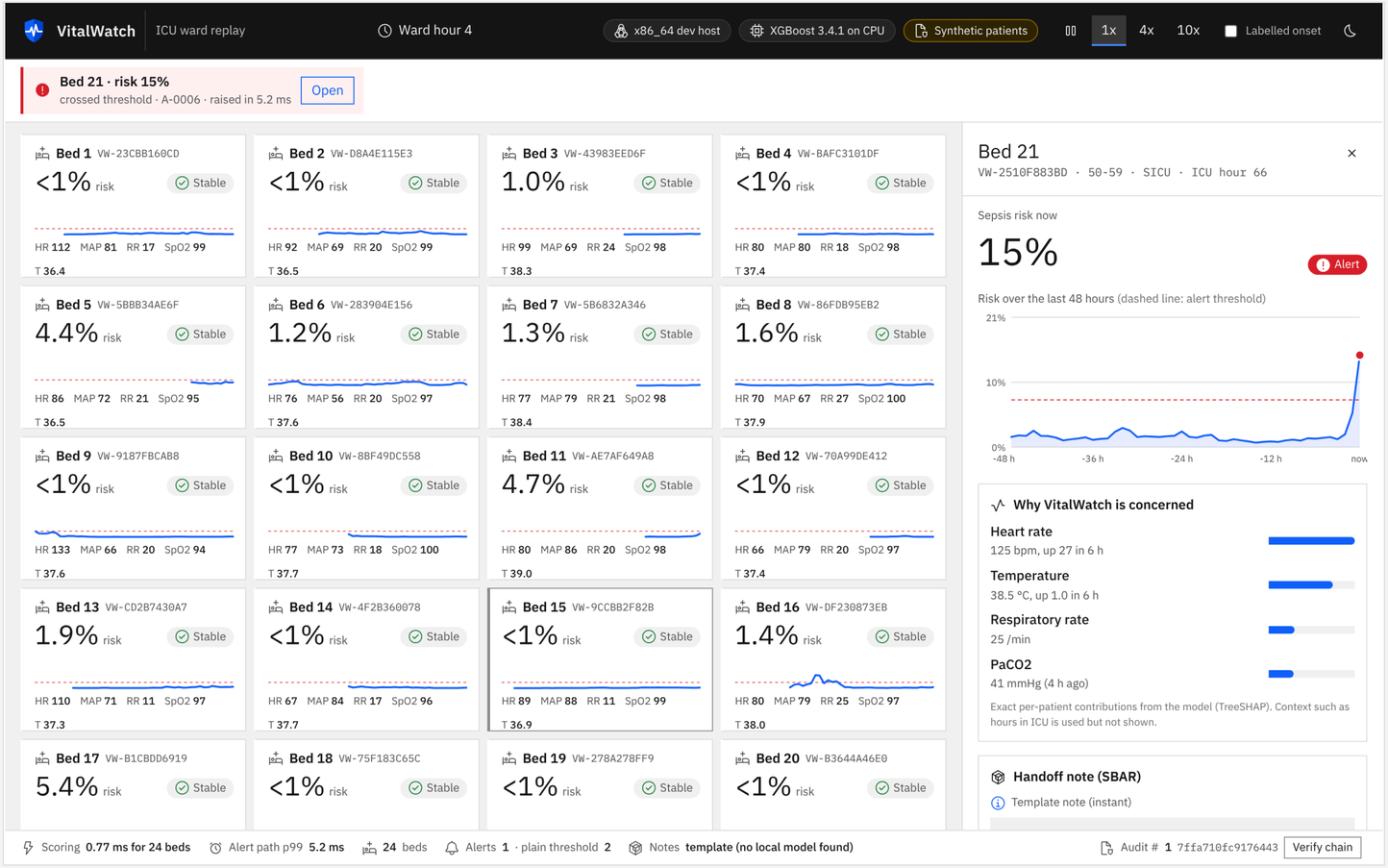
Exact reasons in clinical words, and an SBAR handoff note drafted by IBM Granite.



Private and audited

Granite runs locally; pseudonyms only; every decision sealed in a tamper-evident chain.

One screen for the whole ward



Alert in milliseconds

Bed 21 crossed the alert threshold; the alert was raised 5 ms after its new hourly data arrived.

Why, at a glance

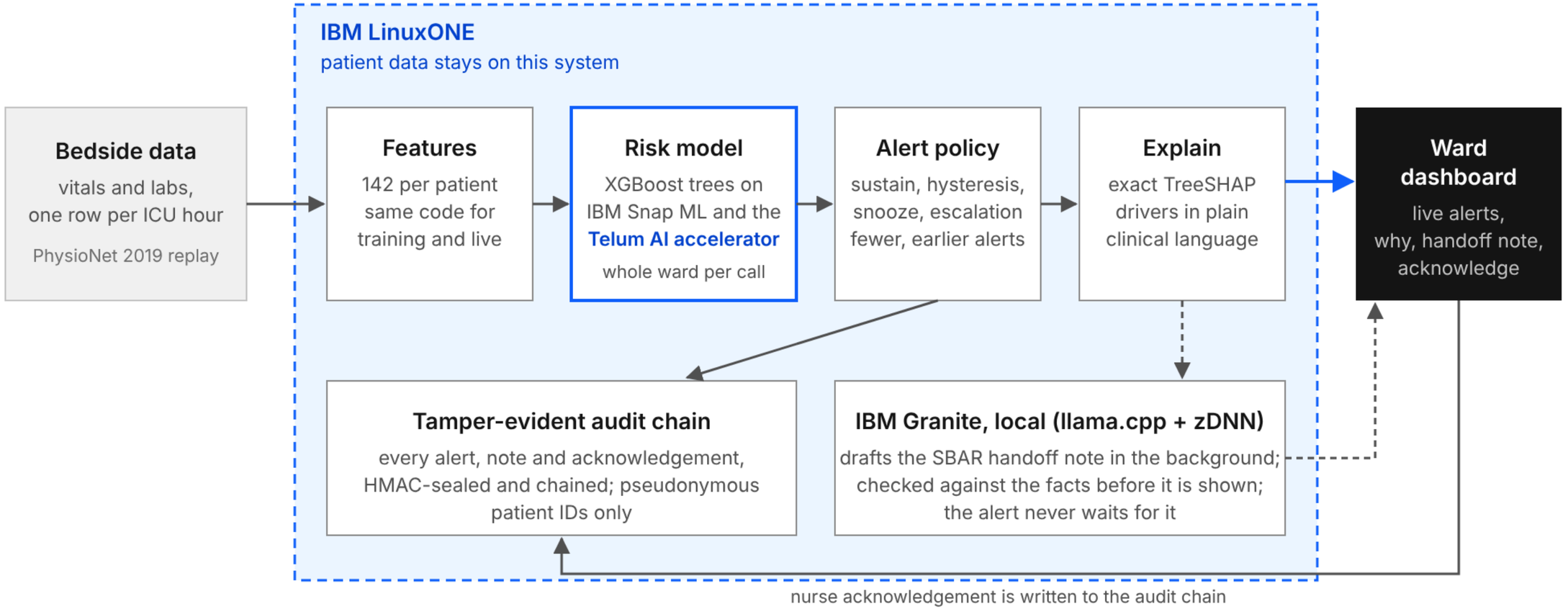
Heart rate 125, up 27 in 6 h; temperature 38.5 °C; respiratory rate 25.

Act and record

Acknowledge with a snooze; the action is sealed in the audit chain.

Screenshot from the live dashboard (synthetic replay); IBM Carbon design language

How it works: one tick is one ICU hour for the whole ward



Same feature code for training and live serving, verified to 1e-12 by a parity test

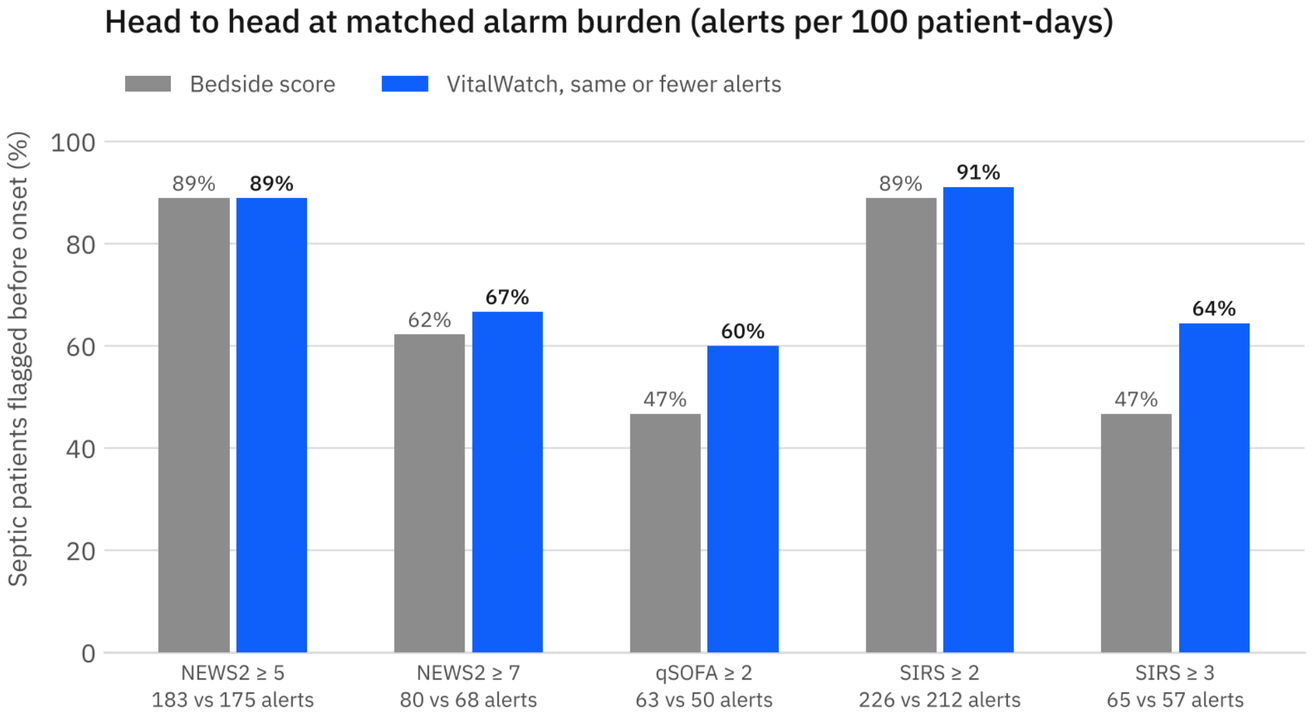
METHOD

Designed to be judged fairly

Step	What we did
Data	PhysioNet/CinC 2019: about 40,000 ICU stays, hourly vitals and labs, two hospital systems (open access)
Split	By patient: 70% train, 15% validation, 15% test; no patient in two splits
Tuning	Threshold and alert policy chosen on validation only; test used once
Comparison	NEWS2, qSOFA, SIRS through the same snooze logic, at matched alarm burden
Metrics	Flagged before onset, lead time, alerts per 100 patient-days, PPV, AUROC
Fairness	Results by sex, age band and hospital; limits stated in the model card

Onset = first SepsisLabel hour + 6 h; detection window 12 h before to 3 h after onset

More septic patients flagged early, with the same or fewer alarms



Synthetic ICU data generated by VitalWatch (not real patients); held-out test patients.

60% vs 47%

flagged before onset, VitalWatch vs qSOFA at fewer alerts (50 vs 63)

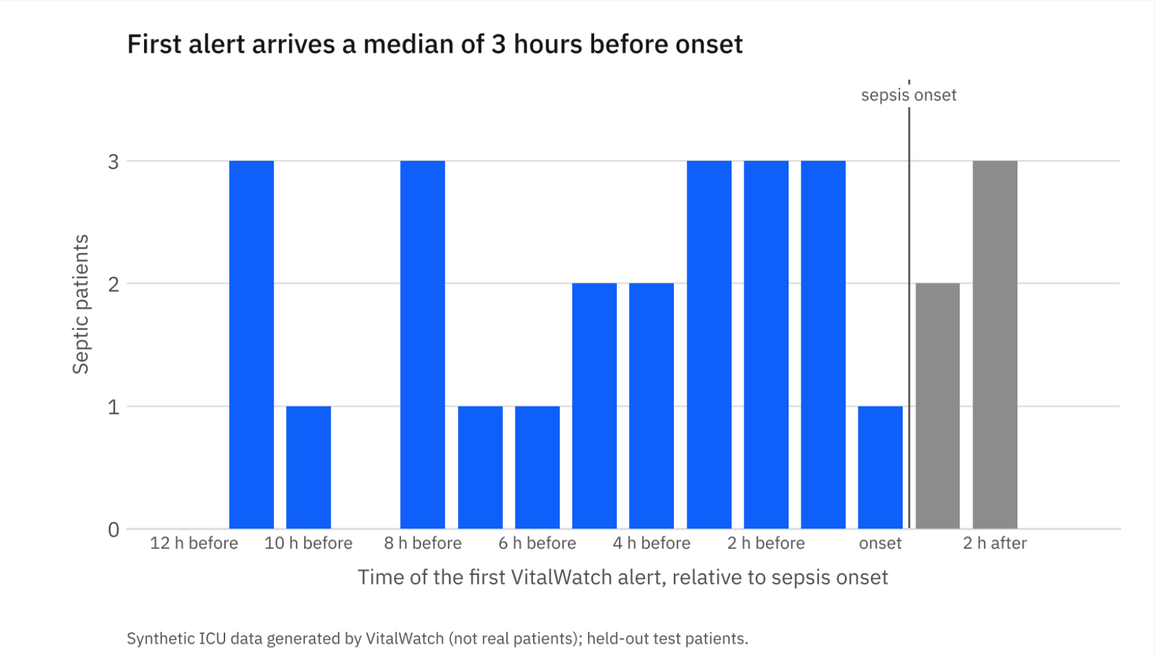
3 h

median warning before onset at 24 alerts per 100 patient-days

Synthetic cohort in the PhysioNet format. The real-data pipeline for PhysioNet 2019 is in the repository.

Held-out test patients; every VitalWatch setting chosen on validation patients

Early, and explained in words a nurse can act on



Handoff note (SBAR), drafted instantly

S: Bed 21, 50-59, SICU, ICU hour 66. Sepsis risk 15%; risk crossed the alert threshold.
B: Heart rate 125 bpm, up 27 in 6 h; temperature 38.5 °C; respiratory rate 25 /min.
A: Possible early deterioration. Decision support, not a diagnosis.
R: Bedside review now; follow the unit sepsis protocol.

IBM Granite, running locally, drafts a richer note in the background. It is shown only if it passes checks: SBAR format, numbers present in the facts, no diagnosis and no drug advice.

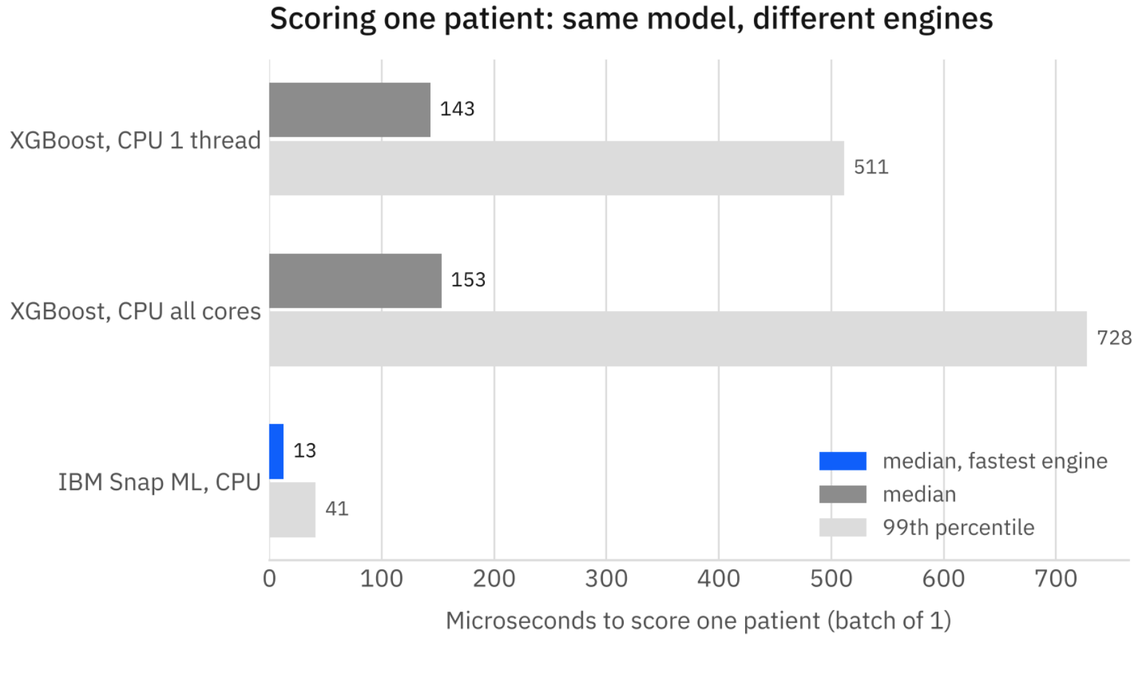
Lead-time chart: synthetic held-out test patients. Note abridged from the demo.

Why IBM Z: AI next to the data, in milliseconds

VitalWatch needs	IBM Z / LinuxONE provides	In this project
Score every bed within milliseconds	Telum on-chip AI accelerator via IBM Snap ML, same XGBoost trees	Built; CPU path measured, Telum path runs on LinuxONE
Generative AI without a cloud API	IBM Granite on llama.cpp with IBM zDNN, local	Built
Same answers on every platform	256 golden predictions re-checked on big-endian s390x	Built; check runs on LinuxONE
Whole-hospital scale on one system	Batched ward scoring and vertical scale	Load-tested to 10,000 beds
Trustworthy records	Tamper-evident chain; pervasive encryption, Crypto Express keys	Chain built; keys: plan
Confidential computing	IBM Secure Execution for Linux	Production plan

Runs in IBM’s stock LinuxONE JupyterLab container plus two packages (Flask, Snap ML)

Same model, faster engine, identical decisions



11x

faster single-patient scoring with IBM Snap ML than XGBoost (13 µs vs 143 µs, x86 reference)

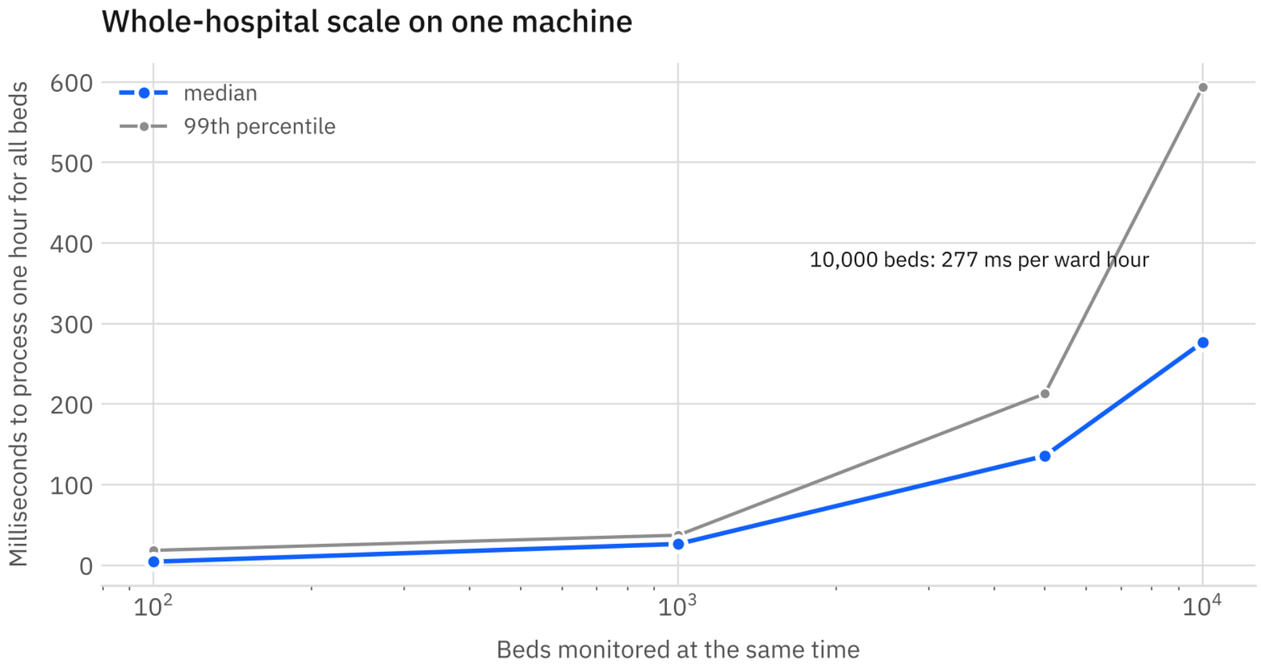
100%

identical alert decisions across engines

On LinuxONE, Snap ML moves the trees onto the Telum AI accelerator. The same bench command adds that row on IBM Z.

Measured with python -m vitalwatch bench; unavailable engines are reported, never estimated

From one ward to a whole health system



x86_64 development machine (4 cores), not IBM Z. Includes features, scoring, policy, explanations and audit.

10,000 beds

processed in 277 ms per ward hour (median), including features, scoring, policy, explanations and audit

35 ms

to score all 10,000 beds in one batched call

Measured on one x86 process with 4 cores, as a reference point, not IBM Z.

Measured with python -m vitalwatch loadtest (100 to 10,000 simultaneous beds)

AI SECURED

Innovation without exposure

Pseudonyms only

Keyed HMAC IDs and age bands on screen, in logs and in prompts. A test fails if a raw ID ever leaks.

Local generative AI

IBM Granite runs on the same LinuxONE system and gets only the minimum facts. No external API call.

Tamper-evident audit

Every alert, note and acknowledgement is HMAC-sealed and chained. Verification names the first altered record.

Production path on IBM Z: Secure Execution for Linux, pervasive encryption with protected keys, and keys held in Crypto Express.

Responsible AI: measured, explained, honest about limits

Flagged before onset, by group

Group	Flagged early
Female / male	50% / 52%
Age 18-49 / 50-69	50% / 63%
Age 70+	38% (monitor)
Hospital A / B	48% / 54%

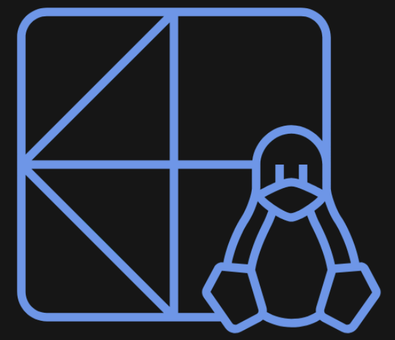
Synthetic held-out test patients; the same table is produced for PhysioNet 2019 by the included pipeline

Stated limits

- Decision support, not a diagnosis or a medical device
- Retrospective data from two hospital systems; prospective validation needed
- Lead time is measured against a retrospective Sepsis-3 label
- Context such as hours in ICU is used but never shown as a reason

Full details: docs/MODEL_CARD.md and docs/PRIVACY.md in the repository

Earlier, fewer, explained alerts, without exposing patient data



NEXT STEPS

Prospective silent-mode trial with a hospital partner · HL7 FHIR integration · Secure Execution deployment · per-unit calibration

github.com/Ritesh-Root/BM-Z-Datathon-2026

Thank you · VitalWatch · IBM Z Datathon 2026 · SSF2026 · built from scratch during the datathon